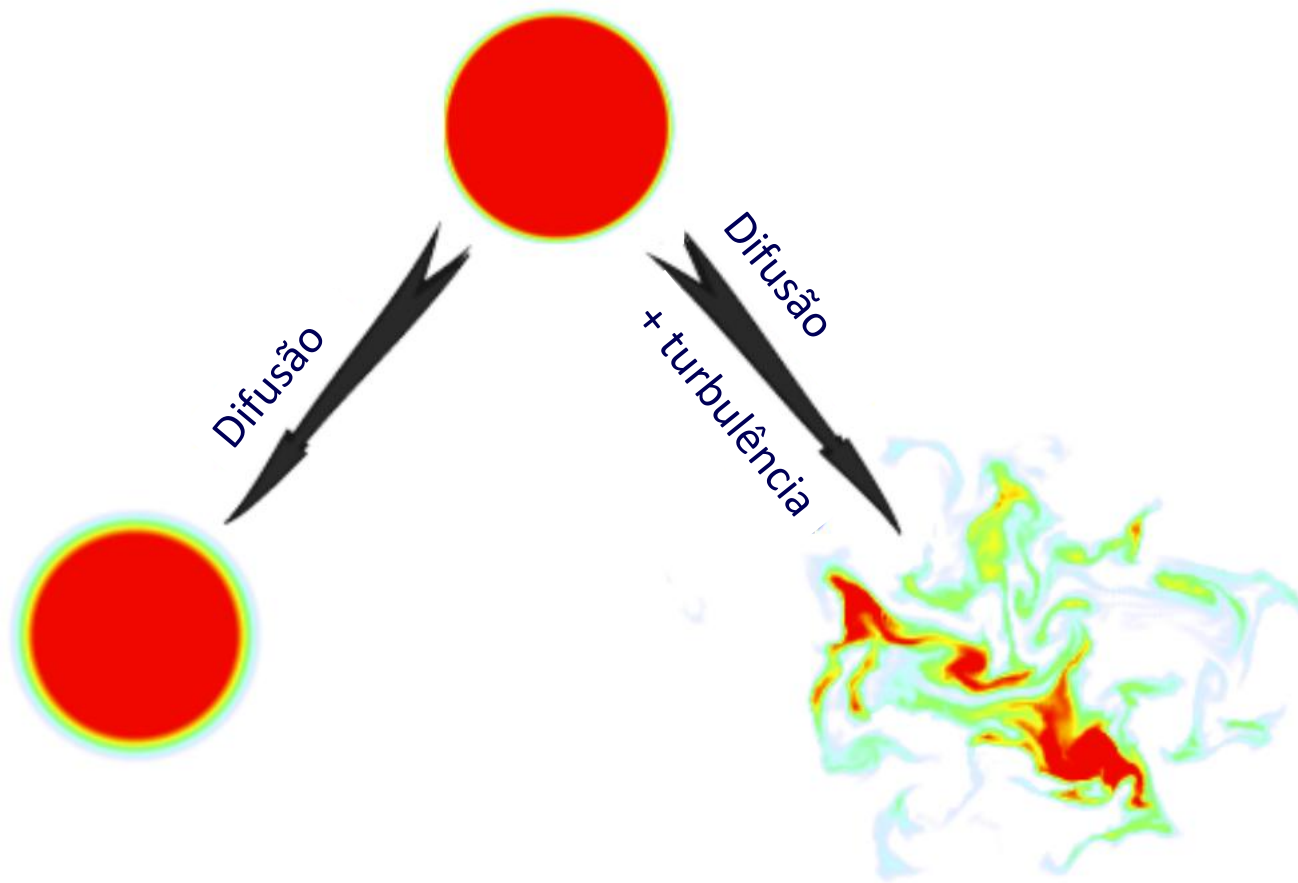


MONITORAMENTO AMBIENTAL SENSORES E MICROCONTROLADORES

AULA 7



Difusão turbulenta



$$\frac{\partial \bar{C}}{\partial t} + \frac{\partial \bar{C} \bar{u}_i}{\partial x_i} = D_m \frac{\partial^2 \bar{C}}{\partial x_i^2} - \frac{\partial \overline{C' u'_i}}{\partial x_i}$$

Coeficiente
de difusão

Contribuição
turbulenta

$$\overline{C' u'_i} = -K \frac{\partial \bar{C}}{\partial x_i}$$

Coeficiente de difusão molecular

$$D \approx \mathcal{O}\left(10^{-9} \frac{\text{m}^2}{\text{s}}\right) \quad \text{Constituintes na água}$$

$$D \approx \mathcal{O}\left(10^{-5} \frac{\text{m}^2}{\text{s}}\right) \quad \text{Constituintes no ar}$$

Coeficiente de difusão turbulenta

$$\text{Rios} \quad K \approx \mathcal{O}\left(10^0 \frac{\text{m}^2}{\text{s}}\right)$$

$$\text{Lagos} \quad K \approx \mathcal{O}\left(10^{-2} \frac{\text{m}^2}{\text{s}}\right)$$

Table 1
Commonly used empirical equations for predicting D_x .

No.	Empirical equations
EF(1)	$D_x = 0.011 U^2 B^2 / (H u_*)$ (Fischer, 1967)
EF(2)	$D_x = 5.93 H u_*$ (Elder, 1959)
EF(3)	$D_x = 0.18 (U/u_*)^{0.5} (B/H)^2 H u_*$ (Liu, 1977)
EF(4)	$D_x = 0.6 (B/H)^2 H u_*$ (Koussis and Rodriguez, 1998)
EF(5)	$D_x = 2.0 (B/H)^{1.5} H u_*$ (Iwasa and Aya, 1991)
EF(6)	$D_x = 0.55 B u_* / H^2$ (Li et al., 1998)
EF(7)	$D_x = 0.2 (U/u_*)^{1.2} (B/H)^{1.3} H u_*$ (Li et al., 1998)
EF(8)	$D_x = 5.92 (U/u_*)^{1.43} (B/H)^{0.62} H u_*$ (Seo and Cheong (1998)
EF(9)	$D_x = 10.612 (U/u_*) H U$ (Kashefipour and Falconer, 2002)
EF(10)	$D_x = (7.428 + 1.775 (B/H)^{0.62} (U/u_*)^{0.572}) (U/u_*) H U$ (Kashefipour and Falconer, 2002)



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Research paper

Estimation of longitudinal dispersion coefficient in rivers

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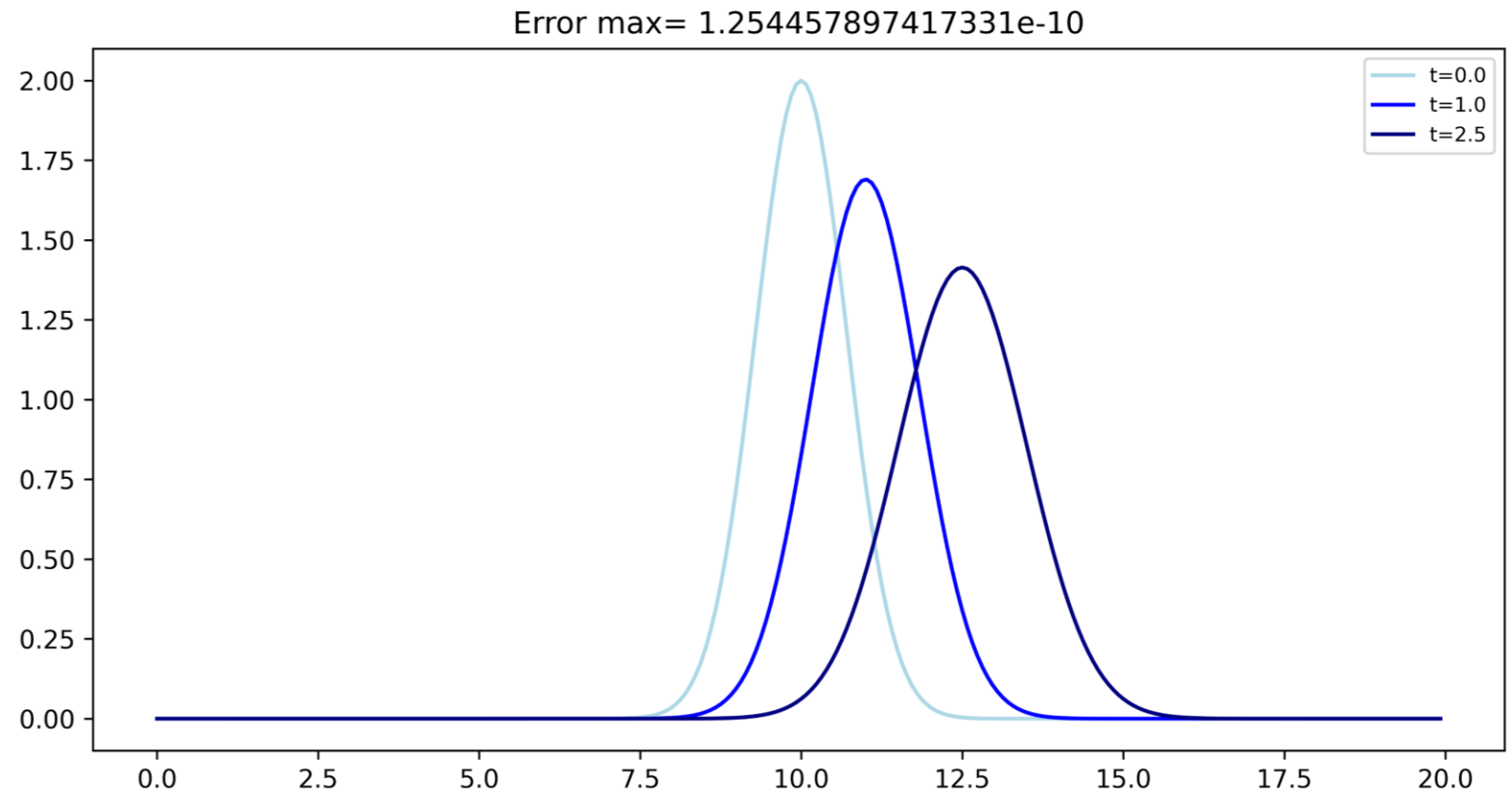
Stream	Dispersion coefficient (m^2/s)
Sabine River, Tex.	12.80
	14.70
	24.20
Mississippi River, La.	237.20
Mississippi River, Mo.	457.70
	374.10
Wind Bighorn River, Wyo.	184.60
	464.60
	16.84
Copper Creep, Va.	14.76
Clinch River, Va.	20.71
Copper Creek, Va.	15.50
Powell River, Tenn.	10.70
Clinch River, Va.	20.82
Copper River, Va.	40.49
Clinch River, Va.	36.93
	24.62
Copper Creek, Va.	1486.45
Missouri River, Iowa	32.52
Bayou Anacoco. La.	39.48
	34.84
Nooksack River, Wash.	41.81
Wind Bighorn River, Wyo.	162.58
	13.94
John Day River, Oreg.	65.03
	111.48
Yadkin River, N.C.	260.13
	22.30
Minnesota River	34.90
	23.20
Amite River	30.20
	30.20
White River	

Stream	Dispersion coefficient (m^2/s)
Nooksack River	153.00
Susquehanna River	92.90
Bayou Anacoco	13.90
Muddy River	13.90
	32.50
Comite River	7.00
	13.90
Missouri River	465.00
	837.00
Missouri River	892.00
Copper Creek, Va.	19.52
	21.40
	9.50
Clinch river, Tn.	13.93
	53.88
	46.45
Copper Creek, Va	9.85
Power River, Tn.	9.50
Clinch River, Va.	8.08
Coachell Canal, Ca.	9.57
Bayou Anacoco	13.94
Nooksake River	153.29
Antietam Creek	9.29
	16.26
	25.55
Monocacy River	4.65
	13.94
	37.16
Missouri River	464.52
	836.13
Wind Bighorn Rivers	41.81
Elkhorn River	9.29

Equação da difusão-advecção

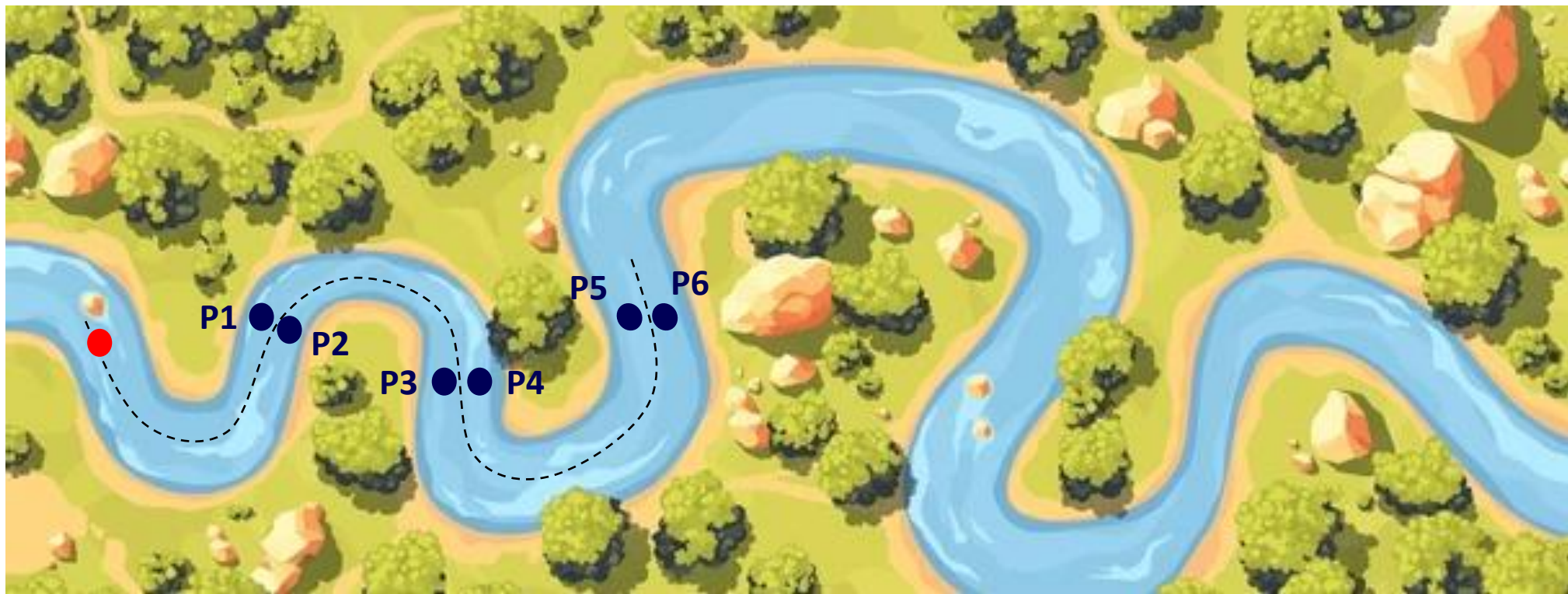
$$\frac{\partial C}{\partial t} + u_{x_i} \frac{\partial C}{\partial x_i} = K_x \frac{\partial^2 C}{\partial x_i^2}$$

$$C(x, t) = \frac{M}{\sqrt{4\pi Kt}} \exp\left(\frac{-(x - ut - x_o)^2}{4Kt}\right)$$



Ponto de lançamento

Ponto de monitoramento





Ponto de monitoramento Condutivímetro desenvolvido

Condutivímetro

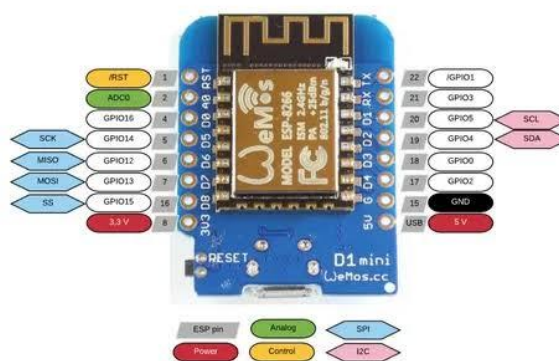
Sensor
DS18B20



Sensor TDS



Wemos



Energia



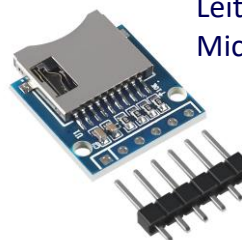
Bateria de lítio de 3.7 V
18650 de 3450 mAh

Botão



Carregador de bateria 5V 1A
18650

Leitor de cartão
MicroSD



Datalogger

Módulo RTC





Ponto de monitoramento Condutivímetro desenvolvido

